THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Expanding Valley Fever Forecasting in California with Environmental LSTMs

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Background

Coccidioidomycosis

An infectious respiratory disease endemic to the southwestern United States. It spreads when soil containing the fungus *Coccidioides* is disturbed and its spores are inhaled.

>20,000 annual cases

Similar to the flu or COVID-19. While symptoms are usually only mild, a small portion of cases are severe, with some even resulting in **death**.

Why Forecast Valley Fever?

Framing the Challenge

Goal: Support earlier outbreak awareness for public health planning.

Case rate data can take up to **2 years** to become publicly available.

Delayed case data limits early outbreak response.

Environmental factors are known to influence when and where case rates rise.

Environmental data is available much sooner.

Research Questions

Without the use of past case rates, can we accurately predict Valley Fever outbreaks across counties before they occur?

Which environmental factors are the strongest drivers of Valley Fever incidence?

How do wildfires influence Valley Fever incidence?

Does larger spatialization allow for generalizable modeling?

Data

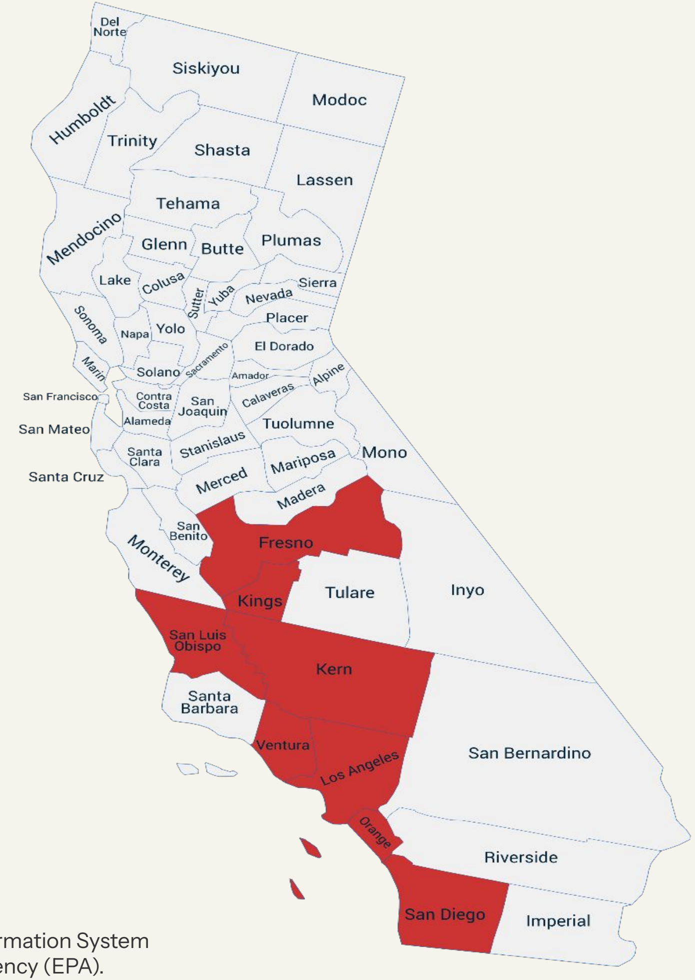
8 counties in California which represent:

- **80.29%** of all CA case rates

Monthly data* from **2001-2024** which include the following environmental features:

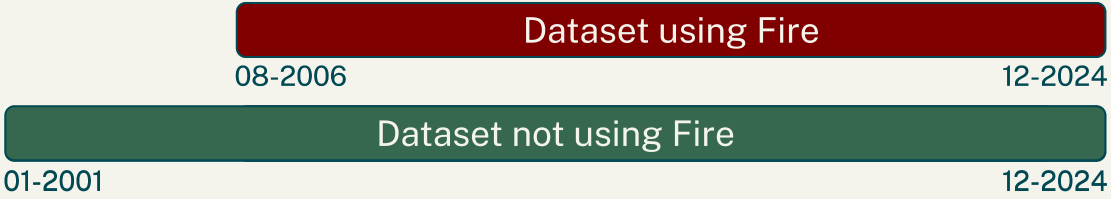
- Precipitation
- Humidity
- Wind Speed
- Wind Run
- Wind Event Count
- Soil Temperature
- Air Quality
- Wildfires
- Fungicide Usage
- Rodenticide Usage

Our target: Valley Fever case rates per 100k



*Data provided by the California Department of Public Health (CDPH), California Irrigation Management Information System (CIMIS), CAL FIRE, California Pesticide Information Portal (CalPIP) and the U.S. Environmental Protection Agency (EPA).

Evaluating the Effectiveness of Wildfire Data

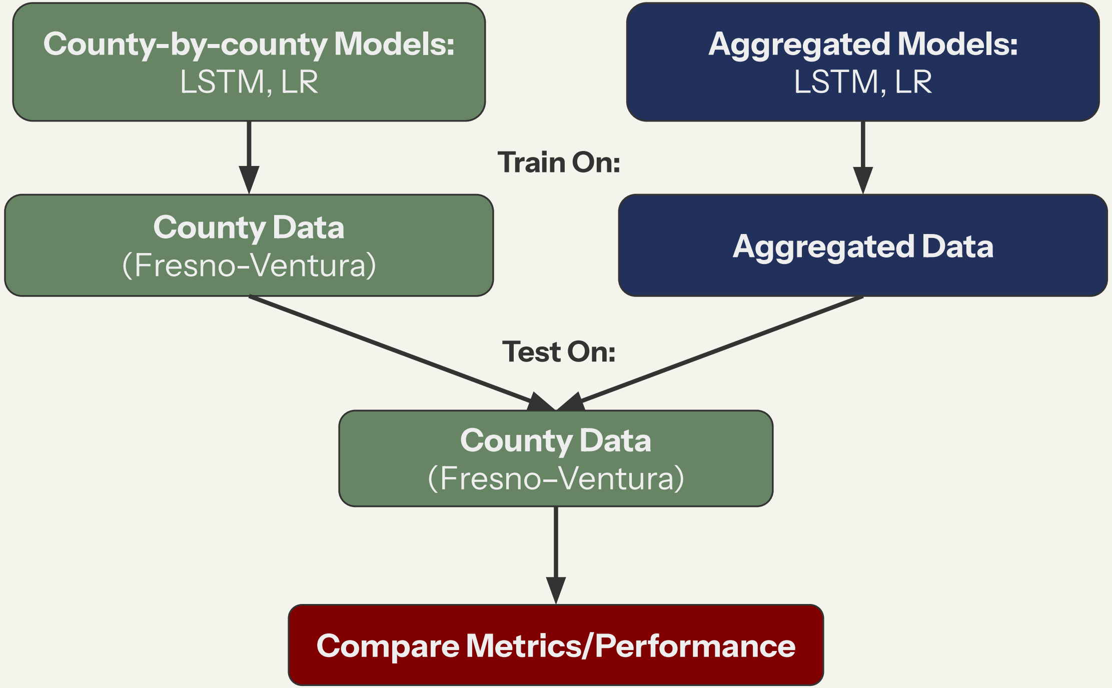


Paired experiment

Each county was modeled twice to isolate the added value of wildfire acres burned.

8 counties × **2 versions: with / without fire** = **16 models**

Methodology

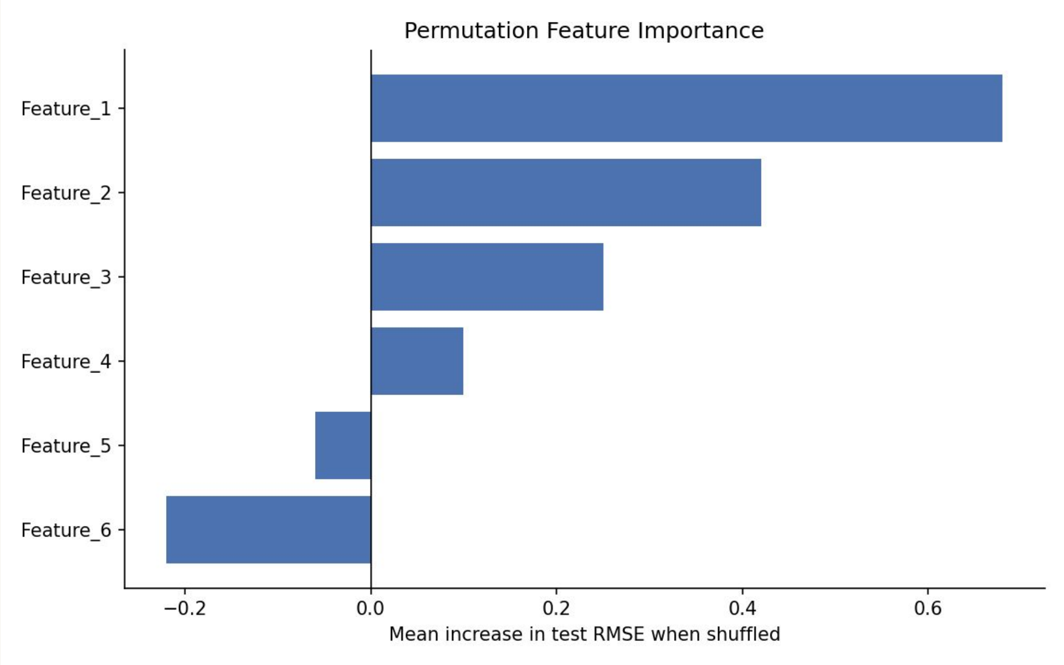


Permutation Feature Importance (PFI)

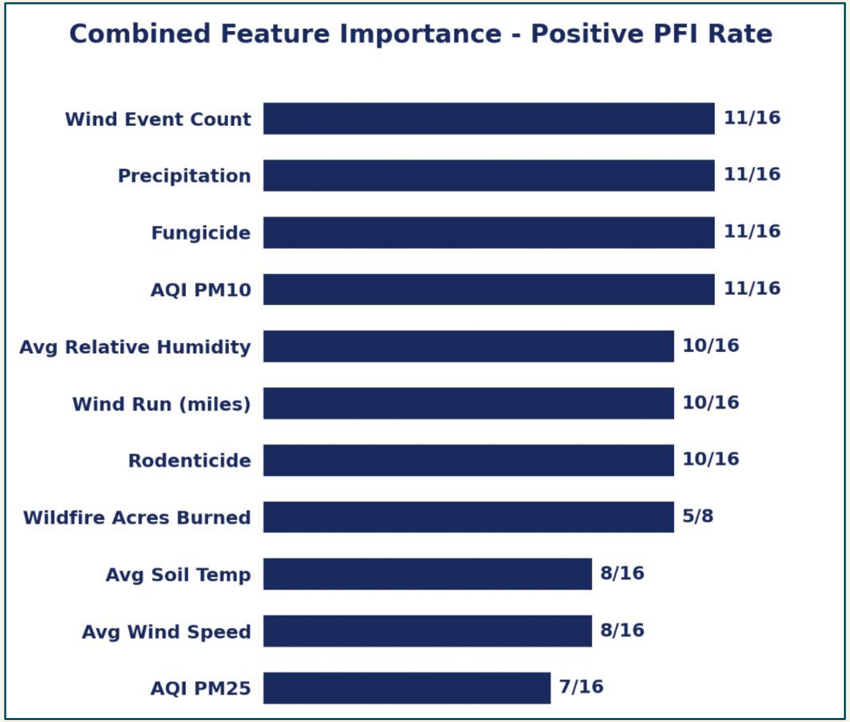
What is PFI?

Permutation Feature Importance is a technique that measures the impact of a variable on a machine learning model.

PFI shuffles one feature at a time in the county validation data and measured the resulting change in RMSE.



Feature Importance Across County Models



- Ran PFI on 8 CA counties, each with/without the fire variable - 16 total models

- Wildfire acres burned as a variable used had marginal improvements in the model

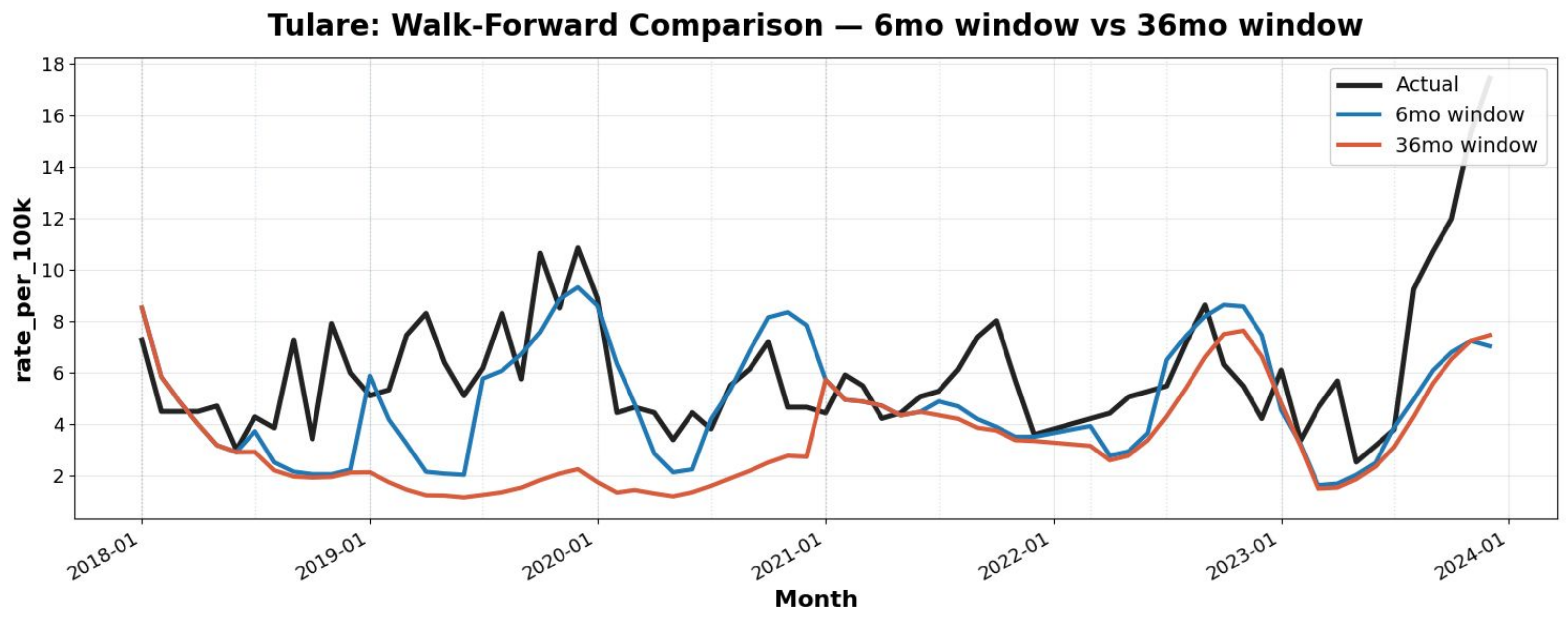
The Leading Environmental Driver Changes by County

- Feature importance varied by county, supporting the use of county-specific feature selection.

- Wind factors appear consistently across counties, reflecting how wind lifts and transports these fungal spores even when it's not a primary predictor.

County	First Highest	Second Highest	Third Highest
Tulare	Soil Temperature	Fungicide Usage	AQI PM25
Kern	Soil Temperature	AQI PM10	Rodenticide Usage
Fresno	Humidity	Soil Temperature	Wind Event Count
Orange	AQI PM25	Precipitation	AQI PM10
Los Angeles	AQI PM10	Wind Event Count	Humidity
San Diego	Precipitation	Humidity	Rodenticide Usage
Ventura	Fungicide Usage	AQI PM10	Rodenticide Usage
San Luis Obispo	Wildfire Acres Burned	Wind Event Count	Fungicide Usage

Shorter windows track recent local shifts more closely



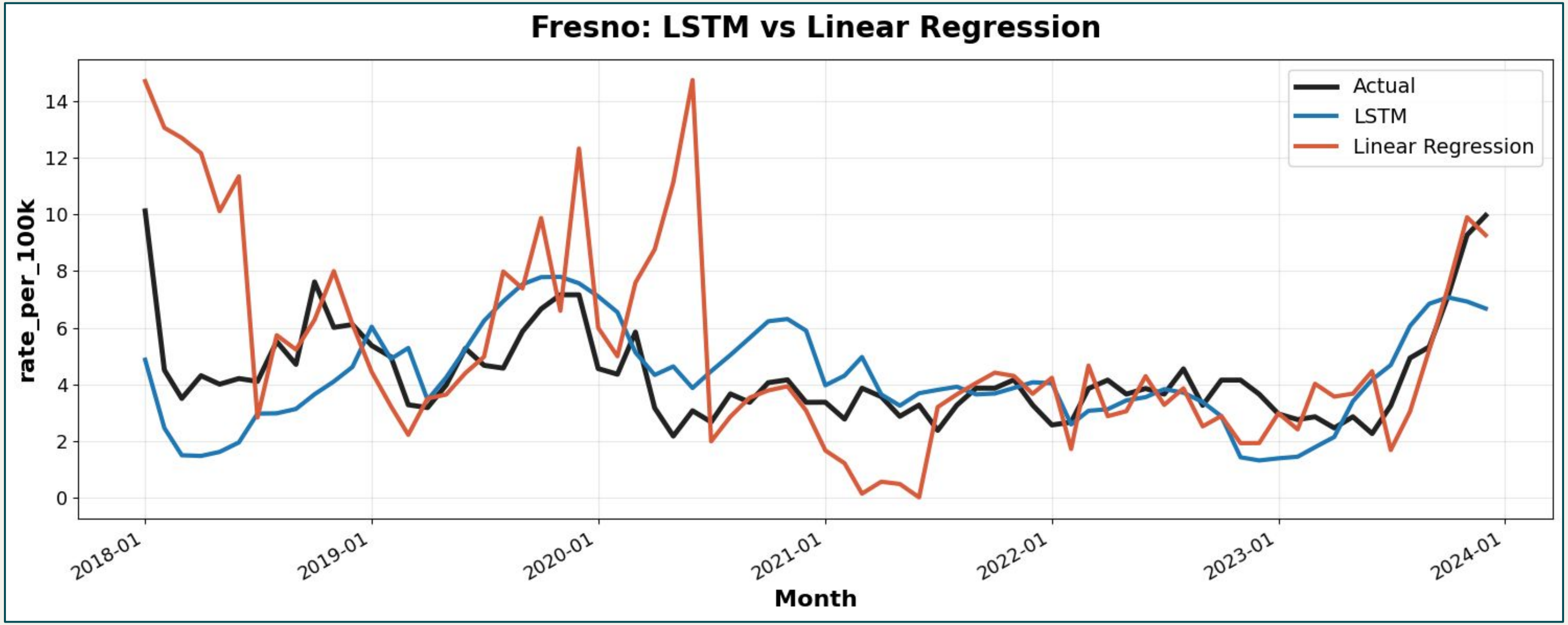
The 6-month window follows recent movement more closely than the 36-month window.

Results

- LSTMs capture dynamics LR misses
- County-by-county LSTMs showed much stronger qualitative performance
- Local Models outperform aggregate models

County	Linear Regression RMSE	Aggregate LSTM RMSE	County by county LSTM RMSE
Fresno	2.737	4.276	**1.607**
Tulare	5.573	3.995	**3.098**
Kern	20.168	24.724	**11.201**
Ventura	**1.853**	3.714	1.967
Orange	**0.234**	2.914	0.276
San Luis Obispo	7.141	7.316	**4.074**
San Diego	**0.384**	2.137	0.558
Los Angeles	**0.214**	2.626	0.309

LSTM captures trends more effectively than linear regression



Conclusions

- Environmental features are effective at providing early warning predictions of Valley Fever.
- Wind events, soil temperature, fungicide use, and PM10 were among the most consistently influential predictors across counties.
- County-specific models and feature sets were more effective than broader aggregated models, suggesting that larger spatialization may obscure important local patterns.
- Wildfire activity was not a consistent county-wide predictor, indicating that its influence may be geographically specific.

These findings can help public health agencies such as the California Department of Public Health focus environmental monitoring by county and employ action before case reports become available.

Future Work

- Create aggregate data using **all** counties
- Group counties into 3 regions:
  - inland, lower coast, and upper coast
- Independent lags for different features
- Fire feature may co-vary with other features, warranting further investigation

Thank You

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